

## 13.7 Where the Form Vanishes

Sections 4 and 5 diagonalized a symmetric form and counted the signs on the diagonal. Over  $\mathbb{R}$  that is the whole story, but it hides the feature that makes indefinite forms geometrically different from inner products: there are non-zero vectors on which the form vanishes. This section takes those vectors seriously. They are not defects to be quarantined; they organize the space, because each one forces a two-dimensional piece of completely standard shape to split off. The result is a decomposition of any non-degenerate space into a pile of identical planes plus a remainder on which the form vanishes nowhere.

Throughout,  $F$  is a field with  $\text{char } F \neq 2$  (Definition 0.7.2),  $V$  is a finite-dimensional vector space over  $F$  with  $\dim V = n$ , and  $\beta$  is a **symmetric** bilinear form on  $V$  (Definition 13.3.1) with quadratic form  $q(\mathbf{v}) = \beta(\mathbf{v}, \mathbf{v})$  (Definition 13.3.2). Two vectors are orthogonal when  $\beta$  pairs them to 0 (Definition 13.4.1). For a subspace  $U \subseteq V$  we write, as in Exercise (C2: Orthogonal complements for a form),

$$U^{\perp\beta} = \{ \mathbf{v} \in V : \beta(\mathbf{u}, \mathbf{v}) = 0 \text{ for every } \mathbf{u} \in U \},$$

and we write  $V = U \perp W$  to say that  $V = U \oplus W$  with every vector of  $U$  orthogonal to every vector of  $W$  — an **orthogonal direct sum**. The subscript  $\beta$  is dropped when only one form is in play.

### 13.7.1 An intuition to give up

Thirteen chapters have trained one reflex above all others:  $\langle \mathbf{v}, \mathbf{v} \rangle = 0$  forces  $\mathbf{v} = \mathbf{0}$ . It is how we proved that an isometry is injective, that an orthogonal list is independent, that a Gram matrix determines its vectors. It is also the first thing that goes.

Take  $V = \mathbb{R}^2$  with

$$q(x_1, x_2) = x_1^2 - x_2^2, \quad \mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The vector  $\mathbf{v} = (1, 1)$  has  $q(\mathbf{v}) = 1 - 1 = 0$ . It is not  $\mathbf{0}$ , and the form is not defective:  $\det \mathbf{A} = -1 \neq 0$ , so  $\beta$  is non-degenerate (Proposition 13.1.2). Indeed  $\beta(\mathbf{v}, (1, -1)) = 1 - (1)(-1) = 2 \neq 0$ , so  $\mathbf{v}$  is perfectly visible to  $\beta$  — it is only invisible to *itself*. The line  $\text{span}(\mathbf{v})$  is a line of “length zero” inside a space where length is otherwise perfectly well behaved.

It gets worse, and the worse case is the one worth internalizing. Take  $V = \mathbb{R}^4$  with

$$q(\mathbf{x}) = x_1^2 + x_2^2 - x_3^2 - x_4^2,$$

of signature  $(2, 2)$  and therefore non-degenerate. Let  $W = \text{span}(\mathbf{e}_1 + \mathbf{e}_3, \mathbf{e}_2 + \mathbf{e}_4)$ , a plane. For  $\mathbf{w} = a(\mathbf{e}_1 + \mathbf{e}_3) + b(\mathbf{e}_2 + \mathbf{e}_4)$ ,

$$q(\mathbf{w}) = a^2 + b^2 - a^2 - b^2 = 0,$$

and in fact  $\beta$  vanishes on every pair from  $W$ , since  $\beta(\mathbf{e}_1 + \mathbf{e}_3, \mathbf{e}_2 + \mathbf{e}_4) = 0$  too. So  $\beta$  restricted to  $W$  is the **zero form** on a two-dimensional subspace of a four-dimensional space on which  $\beta$  is non-degenerate.

Say that again, because it is the sentence an inner-product-trained reader will refuse. A non-degenerate form can vanish identically on a subspace of half the dimension of the whole space. Non-degeneracy is a statement about  $V$ ; it says that no vector is orthogonal to *everything*. It

says nothing whatsoever about what  $\beta$  does inside a subspace. The vectors of  $W$  above are orthogonal to each other and to themselves; they are not orthogonal to  $\mathbf{e}_1 - \mathbf{e}_3$ , and that is all non-degeneracy ever promised.

Once that is accepted, the rest of the section is straightforward. The vanishing vectors get a name, and then a structure theorem.

## 13.7.2 Isotropic vectors and totally isotropic subspaces

*An isotropic vector is a non-zero vector that the form cannot tell from zero when it looks at it twice.*

### Definition 13.7.1 — Isotropic and Anisotropic

Let  $\beta$  be a symmetric bilinear form on  $V$  with quadratic form  $q$ .

- (a) A vector  $\mathbf{v} \in V$  is **isotropic** if  $\mathbf{v} \neq \mathbf{0}$  and  $q(\mathbf{v}) = 0$ , and **anisotropic** if  $q(\mathbf{v}) \neq 0$ .
- (b) The form  $\beta$  is **anisotropic** if it has no isotropic vectors: that is, if  $q(\mathbf{v}) = 0$  implies  $\mathbf{v} = \mathbf{0}$ . Otherwise  $\beta$  is **isotropic**.

Both clauses hinge on small words. In (a) the vector is required to be **non-zero**, so  $\mathbf{0}$  is neither isotropic nor anisotropic; the zero vector always has  $q(\mathbf{0}) = 0$  and carries no information. In (b), “anisotropic” is a property of the **form on a particular space**. Anisotropy passes down to subspaces, since a subspace has fewer vectors to test; isotropy does not, and a form isotropic on  $V$  is very often anisotropic on a subspace. The word comes from *iso* (equal) and *tropos* (direction): a vector along which the form does not distinguish one direction from another.

Examples, over several fields.

- **A positive definite form has no isotropic vectors.** If  $q(\mathbf{v}) > 0$  for  $\mathbf{v} \neq \mathbf{0}$  then  $q(\mathbf{v}) = 0$  forces  $\mathbf{v} = \mathbf{0}$ . So every inner product is anisotropic, which is exactly the reflex described above. The same holds for negative definite forms.
- **Over  $\mathbb{R}$ , signature  $(1, 1)$ .** For  $q = x_1^2 - x_2^2$  the isotropic vectors are the non-zero multiples of  $(1, 1)$  and of  $(1, -1)$ , since  $q = (x_1 - x_2)(x_1 + x_2)$ . Two lines, and nothing else.
- **Over  $\mathbb{C}$ .** For  $q = x_1^2 + x_2^2$  on  $\mathbb{C}^2$  the factorization  $q = (x_1 + ix_2)(x_1 - ix_2)$  is available, so  $(1, i)$  and  $(1, -i)$  are isotropic. Over  $\mathbb{R}$  the same  $q$  is positive definite. **Isotropy depends on the field**, not only on the polynomial.
- **Over  $\mathbb{F}_3$  and  $\mathbb{F}_5$ .** For  $q = x_1^2 + x_2^2$  the squares in  $\mathbb{F}_3$  are 0 and 1, so  $q(\mathbf{x}) = 0$  needs  $x_1 = x_2 = 0$  and the form is anisotropic. In  $\mathbb{F}_5$ , however,  $1^2 + 2^2 = 5 = 0$ , so  $(1, 2)$  is isotropic. The same formula, two finite fields, two answers.
- **The degenerate case.** Every non-zero vector of the radical is isotropic, since  $\mathbf{v} \in \text{rad}(\beta)$  gives  $\beta(\mathbf{u}, \mathbf{v}) = 0$  for every  $\mathbf{u}$ , in particular for  $\mathbf{u} = \mathbf{v}$ . So a degenerate form is automatically isotropic. The converse fails, and the  $\mathbb{R}^2$  example above is the witness.

When a whole subspace consists of isotropic vectors, more is true than the definition says, and it deserves its own name.

### Definition 13.7.2 — Totally Isotropic Subspace

A subspace  $W \subseteq V$  is **totally isotropic** for  $\beta$  if  $\beta(\mathbf{u}, \mathbf{w}) = 0$  for **all**  $\mathbf{u}, \mathbf{w} \in W$ ; equivalently, if  $W \subseteq W^{\perp\beta}$ . The zero subspace is totally isotropic, trivially.

The two formulations agree because  $W \subseteq W^{\perp\beta}$  says exactly that every  $\mathbf{w} \in W$  is orthogonal to every  $\mathbf{u} \in W$ . A third formulation is available in characteristic  $\neq 2$  and is the one to use in practice:  $W$  is totally isotropic if and only if  $q(\mathbf{w}) = 0$  for every  $\mathbf{w} \in W$ . One direction is immediate; the other is Theorem 13.3.5 applied to the restricted form, which recovers  $\beta$  on  $W$  from  $q$  on  $W$  and so makes it zero. In the  $\mathbb{R}^4$  example above we checked exactly this:  $q$  vanished on  $W$ , and therefore so did  $\beta$ .

Totally isotropic subspaces cannot be too large.

### Proposition 13.7.1 — Half the Dimension, at Most

Let  $\beta$  be a non-degenerate symmetric form on  $V$  with  $\dim V = n$ , and let  $W \subseteq V$  be totally isotropic. Then  $\dim W \leq n/2$ .

*Proof.* Since  $\beta$  is non-degenerate,  $\dim W^{\perp\beta} = n - \dim W$  by Exercise (C2: Orthogonal complements for a form) (b). Since  $W$  is totally isotropic,  $W \subseteq W^{\perp\beta}$ , so  $\dim W \leq n - \dim W$ , which is the claim. ■

**Warning.** An isotropic vector is not a zero vector, and a form that vanishes on a subspace need not be degenerate. These are two different errors and both are common. The first confuses  $q(\mathbf{v}) = 0$  with  $\mathbf{v} = \mathbf{0}$ ; over  $\mathbb{R}^2$  with  $q = x_1^2 - x_2^2$  the vector  $(1, 1)$  separates them. The second confuses “ $\beta$  vanishes on  $W \times W$ ” with “ $\beta$  vanishes on  $\{\mathbf{v}\} \times V$ ”. Degeneracy is about the **second**: the radical consists of vectors killed by pairing with *everything*. The plane  $W$  of the  $\mathbb{R}^4$  example carries the zero form and contains no radical vector at all.

*Quick check.* Is the plane  $\text{span}(\mathbf{e}_1, \mathbf{e}_2) \subseteq \mathbb{R}^4$  totally isotropic for  $q(\mathbf{x}) = x_1^2 + x_2^2 - x_3^2 - x_4^2$ ? What about  $\text{span}(\mathbf{e}_1 + \mathbf{e}_3, \mathbf{e}_1 - \mathbf{e}_3)$ ?

*Solution.* No to both. On  $\text{span}(\mathbf{e}_1, \mathbf{e}_2)$  the form restricts to  $x_1^2 + x_2^2$ , which is positive definite, so that plane is as far from totally isotropic as possible: it is anisotropic. The second plane is  $\text{span}(\mathbf{e}_1, \mathbf{e}_3)$ , on which  $q$  restricts to  $x_1^2 - x_3^2$ ; its two spanning vectors  $\mathbf{e}_1 \pm \mathbf{e}_3$  are indeed isotropic, but  $\beta(\mathbf{e}_1 + \mathbf{e}_3, \mathbf{e}_1 - \mathbf{e}_3) = 1 - (-1) = 2 \neq 0$ , so the plane is not totally isotropic. Containing isotropic vectors is not the same as being totally isotropic.

## 13.7.3 Splitting off a non-degenerate piece

Everything below rests on one mechanism: if a subspace carries a non-degenerate restriction, it splits off orthogonally and the rest of the space is again non-degenerate. That is the analogue

of “restrict to an invariant complement”, and it is what makes induction possible.

### Proposition 13.7.2 — A Non-degenerate Subspace Splits Off

Let  $\beta$  be a non-degenerate symmetric form on  $V$  with  $\dim V = n$ , and let  $U \subseteq V$  be a subspace such that  $\beta|_{U \times U}$  is non-degenerate. Then

$$V = U \perp U^{\perp\beta},$$

and  $\beta$  restricted to  $U^{\perp\beta}$  is non-degenerate.

*Idea.* Two things to check and one dimension count. The intersection  $U \cap U^{\perp\beta}$  is precisely the radical of the restricted form, which the hypothesis kills; the dimension of  $U^{\perp\beta}$  is  $n - \dim U$  because the global form is non-degenerate; and a vector of  $U^{\perp\beta}$  orthogonal to  $U^{\perp\beta}$  is orthogonal to all of  $V$ , hence zero.

*Proof.* Write  $U^\perp = U^{\perp\beta}$ . A vector  $\mathbf{v} \in U \cap U^\perp$  lies in  $U$  and satisfies  $\beta(\mathbf{u}, \mathbf{v}) = 0$  for every  $\mathbf{u} \in U$ , so  $\mathbf{v} \in \text{rad}(\beta|_{U \times U})$ , which is  $\{\mathbf{0}\}$  by hypothesis (Definition 13.1.4). Hence  $U \cap U^\perp = \{\mathbf{0}\}$  and the sum  $U + U^\perp$  is direct (Theorem 1.7.4).

Since  $\beta$  is non-degenerate on  $V$ , Exercise (C2: Orthogonal complements for a form) (b) gives  $\dim U^\perp = n - \dim U$ . So  $\dim(U \oplus U^\perp) = n$ , and a subspace of  $V$  of dimension  $n$  is  $V$  (Theorem 1.6.8). The sum is orthogonal by the definition of  $U^\perp$  together with the symmetry of  $\beta$ , so  $V = U \perp U^\perp$ .

Finally let  $\mathbf{w} \in U^\perp$  satisfy  $\beta(\mathbf{x}, \mathbf{w}) = 0$  for every  $\mathbf{x} \in U^\perp$ . For  $\mathbf{u} \in U$  we also have  $\beta(\mathbf{u}, \mathbf{w}) = 0$ , since  $\mathbf{w} \in U^\perp$ . Every  $\mathbf{v} \in V$  is  $\mathbf{u} + \mathbf{x}$  with  $\mathbf{u} \in U$  and  $\mathbf{x} \in U^\perp$ , so  $\beta(\mathbf{v}, \mathbf{w}) = 0$  for every  $\mathbf{v} \in V$ , that is  $\mathbf{w} \in \text{rad}(\beta) = \{\mathbf{0}\}$ . This proves the proposition. ■

## 13.7.4 Hyperbolic planes

Which two-dimensional forms are isotropic? Over  $\mathbb{R}$  with signature  $(1, 1)$  we found two isotropic lines, and the example  $q = x_1^2 - x_2^2$  can be rewritten by the substitution  $u = x_1 + x_2$ ,  $w = (x_1 - x_2)/2$  as  $q = 2uw$  — the cross term alone, with no squares at all. That shape is the model, and it is the same over every field.

### Definition 13.7.3 — Hyperbolic Pair, Hyperbolic Plane

Let  $\beta$  be a symmetric bilinear form on  $V$ . A **hyperbolic pair** is an ordered pair  $(\mathbf{e}, \mathbf{f})$  of vectors of  $V$  with

$$q(\mathbf{e}) = 0, \quad q(\mathbf{f}) = 0, \quad \beta(\mathbf{e}, \mathbf{f}) = 1.$$

The subspace  $H = \text{span}(\mathbf{e}, \mathbf{f})$  is then called a **hyperbolic plane**, and  $(\mathbf{e}, \mathbf{f})$  is a **hyperbolic basis** of it.

Four things follow at once. First,  $\mathbf{e}$  and  $\mathbf{f}$  are linearly independent: if  $\mathbf{f} = c\mathbf{e}$  then  $\beta(\mathbf{e}, \mathbf{f}) =$

$cq(\mathbf{e}) = 0 \neq 1$ . So  $H$  really is a plane,  $\dim H = 2$ . Second, in the basis  $(\mathbf{e}, \mathbf{f})$  the Gram matrix of  $\beta|_H$  is

$$[\beta|_H]_{(\mathbf{e}, \mathbf{f})} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

whose determinant is  $-1 \neq 0$ , so  $\beta|_H$  is **non-degenerate** (Proposition 13.1.2) and Proposition 13.7.2 applies to it. Third,  $q(a\mathbf{e} + b\mathbf{f}) = 2ab$ , so on a hyperbolic plane  $q$  takes **every** value in  $F$ : given  $c \in F$ , take  $a = c/2$  and  $b = 1$ . Fourth, over  $\mathbb{R}$  the Gram matrix has eigenvalues 1 and  $-1$ , so a hyperbolic plane has signature  $(1, 1)$  and is the unique real example up to congruence.

Now the workhorse of this section and the next. It says that one isotropic vector is enough to manufacture a whole hyperbolic plane around it.

### Lemma 13.7.3 — Every Isotropic Vector Lies in a Hyperbolic Plane

Let  $\text{char } F \neq 2$ , let  $\beta$  be a **non-degenerate** symmetric bilinear form on  $V$ , and let  $\mathbf{e} \in V$  be isotropic. Then there is  $\mathbf{f} \in V$  such that  $(\mathbf{e}, \mathbf{f})$  is a hyperbolic pair. In particular  $\mathbf{e}$  lies in a hyperbolic plane  $H = \text{span}(\mathbf{e}, \mathbf{f})$ , and  $\beta|_H$  is non-degenerate.

*Idea.* Non-degeneracy hands us some  $\mathbf{u}$  with  $\beta(\mathbf{e}, \mathbf{u}) \neq 0$ , and rescaling makes  $\beta(\mathbf{e}, \mathbf{u}) = 1$ . Only one condition is still missing,  $q(\mathbf{u}) = 0$ , and  $\mathbf{u}$  can be corrected without disturbing what has been achieved: adding a multiple of  $\mathbf{e}$  changes  $q(\mathbf{u})$  but not  $\beta(\mathbf{e}, \mathbf{u})$ , because  $q(\mathbf{e}) = 0$ . Solving for the multiple is one line, and it divides by 2 — which is where the characteristic hypothesis is spent, for the third time in this chapter.

*Proof.* Since  $\mathbf{e} \neq \mathbf{0}$  and  $\beta$  is non-degenerate, there is  $\mathbf{u}_0 \in V$  with  $c = \beta(\mathbf{e}, \mathbf{u}_0) \neq 0$  (Definition 13.1.4). Put  $\mathbf{u} = c^{-1}\mathbf{u}_0$ , so that

$$\beta(\mathbf{e}, \mathbf{u}) = 1.$$

Now set

$$\mathbf{f} = \mathbf{u} - \frac{1}{2}q(\mathbf{u})\mathbf{e},$$

which is legitimate because  $\text{char } F \neq 2$  makes  $2 = 1+1$  invertible. Then, using bilinearity, symmetry and  $q(\mathbf{e}) = 0$ ,

$$\beta(\mathbf{e}, \mathbf{f}) = \beta(\mathbf{e}, \mathbf{u}) - \frac{1}{2}q(\mathbf{u})q(\mathbf{e}) = 1 - 0 = 1,$$

and

$$\begin{aligned} q(\mathbf{f}) &= q(\mathbf{u}) - 2 \cdot \frac{1}{2}q(\mathbf{u})\beta(\mathbf{u}, \mathbf{e}) + \frac{1}{4}q(\mathbf{u})^2q(\mathbf{e}) \\ &= q(\mathbf{u}) - q(\mathbf{u}) + 0 = 0. \end{aligned}$$

Together with  $q(\mathbf{e}) = 0$ , this shows that  $(\mathbf{e}, \mathbf{f})$  is a hyperbolic pair. As noted after Definition 13.7.3,  $\mathbf{e}$  and  $\mathbf{f}$  are independent, so  $H = \text{span}(\mathbf{e}, \mathbf{f})$  is a plane, and the Gram matrix  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  is invertible, so  $\beta|_H$  is non-degenerate. This proves the lemma. ■

*Remark.* Non-degeneracy of  $\beta$  is not decoration here. On  $\mathbb{R}^2$  with  $q(\mathbf{x}) = x_1^2$ , the vector  $\mathbf{e}_2$  is isotropic and lies in no hyperbolic plane, because  $\beta(\mathbf{e}_2, \mathbf{v}) = 0$  for every  $\mathbf{v}$  and a hyperbolic pair needs  $\beta(\mathbf{e}, \mathbf{f}) = 1$ . What fails is the very first line of the proof.

## 13.7.5 The Witt decomposition

Peeling off a hyperbolic plane and repeating is now a two-line induction, and it is the structural theorem of the section: it does for indefinite forms what “peel off an eigenvector” did for operators in Chapter 8.

### Theorem 13.7.4 — Witt Decomposition

Let  $\text{char } F \neq 2$  and let  $\beta$  be a non-degenerate symmetric bilinear form on a finite-dimensional  $V$ . Then there are an integer  $m \geq 0$ , hyperbolic planes  $H_1, \dots, H_m \subseteq V$  and a subspace  $V_0 \subseteq V$  such that

$$V = H_1 \perp H_2 \perp \cdots \perp H_m \perp V_0$$

and  $\beta$  restricted to  $V_0$  is **anisotropic**. In particular  $\dim V = 2m + \dim V_0$ .

*Idea.* Induction on  $\dim V$ . Either  $\beta$  is anisotropic, in which case take  $m = 0$  and  $V_0 = V$  and stop; or there is an isotropic vector, in which case [Lemma 13.7.3](#) wraps a hyperbolic plane  $H_1$  around it, [Proposition 13.7.2](#) peels  $H_1$  off with a non-degenerate remainder  $H_1^\perp$  of dimension  $\dim V - 2$ , and the induction hypothesis finishes the job there.

*Proof.* Induct on  $\dim V$ . If  $\dim V = 0$ , take  $m = 0$  and  $V_0 = \{0\}$ , on which the form is anisotropic vacuously. Let  $\dim V \geq 1$  and suppose the statement holds for every non-degenerate symmetric form on a space of smaller dimension.

If  $\beta$  is anisotropic, take  $m = 0$  and  $V_0 = V$ .

Otherwise  $\beta$  has an isotropic vector  $\mathbf{e}$ . By [Lemma 13.7.3](#) there is  $\mathbf{f}$  with  $(\mathbf{e}, \mathbf{f})$  a hyperbolic pair, and  $H_1 = \text{span}(\mathbf{e}, \mathbf{f})$  is a hyperbolic plane with  $\beta|_{H_1}$  non-degenerate. By [Proposition 13.7.2](#),

$$V = H_1 \perp H_1^{\perp\beta},$$

and  $\beta$  restricted to  $V' = H_1^{\perp\beta}$  is non-degenerate, with  $\dim V' = \dim V - 2 < \dim V$ . The induction hypothesis applied to  $(V', \beta|_{V'})$  gives hyperbolic planes  $H_2, \dots, H_m \subseteq V'$  and an anisotropic  $V_0 \subseteq V'$  with  $V' = H_2 \perp \cdots \perp H_m \perp V_0$ . Each  $H_i$  with  $i \geq 2$  and  $V_0$  lie inside  $V' = H_1^{\perp\beta}$ , so all the summands are pairwise orthogonal, and

$$V = H_1 \perp H_2 \perp \cdots \perp H_m \perp V_0.$$

Counting dimensions,  $\dim V = 2m + \dim V_0$ . This completes the induction. ■

Two questions are now unavoidable: is  $m$  the same for every such decomposition, and is  $V_0$  determined? The first has an answer here, and it comes from a quantity that mentions no decomposition at all.

### Definition 13.7.4 — Witt Index

Let  $\beta$  be a non-degenerate symmetric bilinear form on a finite-dimensional  $V$ . The **Witt**

**index** of  $\beta$  is

$$\text{ind}(\beta) := \max\{\dim W : W \subseteq V \text{ totally isotropic}\},$$

the largest dimension of a totally isotropic subspace. By [Proposition 13.7.1](#) it satisfies  $\text{ind}(\beta) \leq n/2$ , and it is 0 exactly when  $\beta$  is anisotropic.

The maximum exists because the dimensions in question form a non-empty finite set of integers:  $W = \{0\}$  is always totally isotropic, and all dimensions are at most  $n$ . Nothing in the definition refers to a basis or a decomposition, so the Witt index is an invariant of  $(V, \beta)$  by construction. It is also exactly the number of planes.

### Proposition 13.7.5 — The Number of Hyperbolic Planes Is the Witt Index

In any decomposition  $V = H_1 \perp \cdots \perp H_m \perp V_0$  as in [Theorem 13.7.4](#),  $m = \text{ind}(\beta)$ . In particular  $m$  is the same for every such decomposition, and so is  $\dim V_0 = n - 2m$ .

*Idea.* One inequality is a construction: the first vectors  $\mathbf{e}_1, \dots, \mathbf{e}_m$  of the  $m$  hyperbolic bases span a totally isotropic subspace of dimension  $m$ , since each is isotropic and different planes are orthogonal. The other inequality is an induction that trades one plane for one dimension: given a totally isotropic  $W$ , the vectors of  $W$  orthogonal to  $\mathbf{e}_1$  project into  $H_1^\perp$  as a totally isotropic subspace of dimension at least  $\dim W - 1$ . Repeating  $m$  times lands inside the anisotropic  $V_0$ , where the only totally isotropic subspace is  $\{0\}$ .

*Proof.* Let  $(\mathbf{e}_i, \mathbf{f}_i)$  be a hyperbolic basis of  $H_i$  for each  $i$ .

**Step 1:**  $\text{ind}(\beta) \geq m$ . Put  $W = \text{span}(\mathbf{e}_1, \dots, \mathbf{e}_m)$ . The  $\mathbf{e}_i$  lie in different summands of a direct sum and are non-zero, so they are independent and  $\dim W = m$ . For  $i \neq j$  we have  $\beta(\mathbf{e}_i, \mathbf{e}_j) = 0$  because  $H_i \perp H_j$ , and  $\beta(\mathbf{e}_i, \mathbf{e}_i) = q(\mathbf{e}_i) = 0$ . By bilinearity  $\beta$  vanishes on  $W \times W$ , so  $W$  is totally isotropic of dimension  $m$ .

**Step 2: a plane can be traded for a dimension.**

*Claim.* Let  $\beta$  be non-degenerate on  $V$ , let  $H = \text{span}(\mathbf{e}, \mathbf{f}) \subseteq V$  be a hyperbolic plane, let  $V' = H^\perp$ , and let  $W \subseteq V$  be totally isotropic. Then  $V'$  contains a totally isotropic subspace of dimension at least  $\dim W - 1$ .

*Proof.* By Proposition 13.7.2,  $V = H \perp V'$ ; let  $P: V \rightarrow V'$  be the projection with kernel  $H$ , so that  $\mathbf{v} - P(\mathbf{v}) \in H$  for every  $\mathbf{v}$ . Put

$$W_1 = \{ \mathbf{w} \in W : \beta(\mathbf{e}, \mathbf{w}) = 0 \},$$

the kernel of a linear functional on  $W$ , so  $\dim W_1 \geq \dim W - 1$ .

For  $\mathbf{w} \in V$  write  $\mathbf{w} - P(\mathbf{w}) = a\mathbf{e} + b\mathbf{f}$ . Since  $\mathbf{e} \in H$  is orthogonal to  $V'$ ,

$$\beta(\mathbf{e}, \mathbf{w}) = \beta(\mathbf{e}, a\mathbf{e} + b\mathbf{f}) = a q(\mathbf{e}) + b = b.$$

So  $\mathbf{w} \in W_1$  forces  $\mathbf{w} - P(\mathbf{w}) \in \text{span}(\mathbf{e})$ . Consequently, for  $\mathbf{w}, \mathbf{w}' \in W_1$  the vectors  $\mathbf{w} - P(\mathbf{w})$  and  $\mathbf{w}' - P(\mathbf{w}')$  both lie in  $\text{span}(\mathbf{e})$ , where  $\beta$  vanishes because  $q(\mathbf{e}) = 0$ ; hence

$$\beta(P(\mathbf{w}), P(\mathbf{w}')) = \beta(\mathbf{w}, \mathbf{w}') - \beta(\mathbf{w} - P(\mathbf{w}), \mathbf{w}' - P(\mathbf{w}')) = 0 - 0 = 0,$$

the first equality using the orthogonality of  $H$  and  $V'$ . So  $P(W_1)$  is a totally isotropic subspace of  $V'$ .

It remains to bound its dimension. By Rank–Nullity (Theorem 3.3.1) applied to  $P|_{W_1}$ , whose kernel is  $W_1 \cap H$ ,

$$\dim P(W_1) = \dim W_1 - \dim(W_1 \cap H).$$

If  $W_1 \cap H = \{0\}$ , then  $\dim P(W_1) = \dim W_1 \geq \dim W - 1$  and we are done. Otherwise pick  $\mathbf{x} \neq 0$  in  $W_1 \cap H$ . Then  $P(\mathbf{x}) = 0$ , so  $\mathbf{x} = \mathbf{x} - P(\mathbf{x}) \in \text{span}(\mathbf{e})$  by the paragraph above, whence  $\mathbf{e} \in W$ . But then  $\beta(\mathbf{e}, \mathbf{w}) = 0$  for every  $\mathbf{w} \in W$ , because  $W$  is totally isotropic and  $\mathbf{e} \in W$ ; so  $W_1 = W$ . Moreover every  $\mathbf{x} \in W_1 \cap H$  lies in  $\text{span}(\mathbf{e})$  by the same argument, so  $W_1 \cap H = \text{span}(\mathbf{e})$  has dimension 1, and  $\dim P(W_1) = \dim W - 1$ . Either way the claim holds. ■

**Step 3:**  $\text{ind}(\beta) \leq m$ . Let  $W \subseteq V$  be totally isotropic. Apply Step 2 to  $H = H_1$ : since  $H_1^{\perp\beta} = H_2 \perp \cdots \perp H_m \perp V_0$  carries a non-degenerate form, we obtain a totally isotropic subspace of it of dimension at least  $\dim W - 1$ . Repeating with  $H_2, \dots, H_m$  in turn produces a totally isotropic subspace of  $V_0$  of dimension at least  $\dim W - m$ . But  $\beta|_{V_0}$  is anisotropic, so a non-zero vector of  $V_0$  has  $q \neq 0$  and the only totally isotropic subspace of  $V_0$  is  $\{0\}$ . Hence  $\dim W - m \leq 0$ .

Steps 1 and 3 give  $\text{ind}(\beta) = m$ , and  $\dim V_0 = n - 2m$  follows by counting. This proves the proposition. ■

So the number of planes is pinned down. What is not yet pinned down is  $V_0$  itself: different decompositions produce different subspaces, and the claim one wants is that they all carry *the same form* in a suitable sense. Making that precise needs the word “isometry” for forms, and proving it needs Witt’s theorems; both are the business of the next section.

## 13.7.6 The two familiar fields

Over  $\mathbb{R}$  and  $\mathbb{C}$  everything can be computed, because the classification is already known.

### Proposition 13.7.6 — The Witt Index over the Reals and the Complexes (a)

Let  $\beta$  be a non-degenerate symmetric form on a real  $V$  with signature  $(n_+, n_-)$  (Definition 13.5.1), so  $n_+ + n_- = n$ . Then  $\text{ind}(\beta) = \min(n_+, n_-)$ , and the anisotropic part has dimension  $|n_+ - n_-|$  and is definite.

- (b) Let  $\beta$  be a non-degenerate symmetric form on a complex  $V$  with  $\dim V = n$ . Then  $\text{ind}(\beta) = \lfloor n/2 \rfloor$ . In particular every such form with  $n \geq 2$  has an isotropic vector, and the anisotropic part has dimension 0 or 1.

*Proof.*

- (a) Put  $k = \min(n_+, n_-)$ . By Theorem 13.4.2 and the rescaling that produced Eq. 13.5.1, together with  $n_0 = 0$ , there is an orthogonal basis  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  with  $q(\mathbf{v}_i) = 1$  for  $i \leq n_+$ ,  $q(\mathbf{v}_i) = -1$  for  $n_+ < i \leq n$ , and distinct basis vectors orthogonal. For  $1 \leq i \leq k$  set

$$\mathbf{e}_i = \mathbf{v}_i + \mathbf{v}_{n_++i}, \quad \mathbf{f}_i = \frac{1}{2}(\mathbf{v}_i - \mathbf{v}_{n_++i}).$$

Then  $q(\mathbf{e}_i) = 1 + (-1) = 0$  and  $q(\mathbf{f}_i) = \frac{1}{4}(1 + (-1)) = 0$ , while

$$\beta(\mathbf{e}_i, \mathbf{f}_i) = \frac{1}{2}(q(\mathbf{v}_i) - q(\mathbf{v}_{n_++i})) = \frac{1}{2}(1 + 1) = 1,$$

so  $(\mathbf{e}_i, \mathbf{f}_i)$  is a hyperbolic pair and  $H_i = \text{span}(\mathbf{e}_i, \mathbf{f}_i)$  a hyperbolic plane. These planes are pairwise orthogonal, since for  $i \neq j$  their spanning vectors involve disjoint sets of the  $\mathbf{v}$ 's. Let  $V_0$  be the span of the remaining  $|n_+ - n_-|$  basis vectors, all of the same sign; then  $V = H_1 \perp \dots \perp H_k \perp V_0$ , and  $\beta|_{V_0}$  is definite, hence anisotropic. By Proposition 13.7.5,  $\text{ind}(\beta) = k$ .

- (b) By Corollary 13.4.4 and non-degeneracy, there is a basis  $(\mathbf{v}_1, \dots, \mathbf{v}_n)$  with  $q(\mathbf{v}_j) = 1$  for every  $j$  and distinct basis vectors orthogonal. Write  $n = 2k + r$  with  $r \in \{0, 1\}$ , and let  $i$  denote the imaginary unit. For  $1 \leq j \leq k$  set

$$\mathbf{e}_j = \mathbf{v}_{2j-1} + i \mathbf{v}_{2j}, \quad \mathbf{f}_j = \frac{1}{2}(\mathbf{v}_{2j-1} - i \mathbf{v}_{2j}).$$

Then  $q(\mathbf{e}_j) = 1 + i^2 = 0$  and likewise  $q(\mathbf{f}_j) = \frac{1}{4}(1 + i^2) = 0$ , while  $\beta(\mathbf{e}_j, \mathbf{f}_j) = \frac{1}{2}(1 + 1) = 1$ . As in (a) these span  $k$  pairwise orthogonal hyperbolic planes, and what is left is the span of  $\mathbf{v}_n$  when  $r = 1$ , on which  $q(c\mathbf{v}_n) = c^2 \neq 0$  for  $c \neq 0$ , and  $\{\mathbf{0}\}$  when  $r = 0$ . So  $\text{ind}(\beta) = k = \lfloor n/2 \rfloor$  by Proposition 13.7.5. This proves the proposition. ■

Two readings are worth recording. Over  $\mathbb{R}$  the Witt index is the smaller of the two counts of signs: a definite form has index 0, and a form of signature  $(m, m)$  has the largest possible index  $m = n/2$  — the  $\mathbb{R}^4$  example that opened the section, with  $n_+ = n_- = 2$ , sits exactly there. Over  $\mathbb{C}$  the index is as large as it can be for every non-degenerate form, because  $-1$  is a square and so a sum of two squares always factors; the only anisotropic complex spaces are  $\{\mathbf{0}\}$  and the lines.

*Quick check.* A non-degenerate real form has signature  $(3, 1)$ . What is its Witt index, and what is the anisotropic part?

*Solution.* The index is  $\min(3, 1) = 1$ , so exactly one hyperbolic plane splits off and  $\dim V_0 = 4 - 2 = 2$ . By [Proposition 13.7.6](#) (a),  $V_0$  carries a definite form of dimension  $|3 - 1| = 2$ , positive definite here since the positive signs are the ones in surplus. Concretely  $\mathbb{R}^{3,1}$  with  $q = x_1^2 + x_2^2 + x_3^2 - x_4^2$  splits as one hyperbolic plane, spanned by  $\mathbf{e}_3 \pm \mathbf{e}_4$  after rescaling, orthogonal to the positive definite plane  $\text{span}(\mathbf{e}_1, \mathbf{e}_2)$ . The largest totally isotropic subspace is a line, and the lines of this kind sweep out the light cone that Section 11 studies.

## 13.7.7 Exercises

### 13.7.7.1 A. Check your understanding

#### Exercise A1.

- Define **isotropic vector**, **totally isotropic subspace** and **hyperbolic pair**.
- Determine whether the following statement is correct, and justify your answer: a non-degenerate form has no isotropic vectors.
- What is the Witt index of a positive definite form on a space of dimension  $n$ ? Of a form of signature  $(4, 4)$ ?
- State the Gram matrix of a hyperbolic plane in a hyperbolic basis, and say why it shows the restricted form is non-degenerate.
- Which hypothesis of [Lemma 13.7.3](#) fails for the form  $q(\mathbf{x}) = x_1^2$  on  $\mathbb{R}^2$  at the isotropic vector  $\mathbf{e}_2$ ?

*Solution.*

- A vector  $\mathbf{v}$  is isotropic if  $\mathbf{v} \neq \mathbf{0}$  and  $q(\mathbf{v}) = 0$ . A subspace  $W$  is totally isotropic if  $\beta(\mathbf{u}, \mathbf{w}) = 0$  for all  $\mathbf{u}, \mathbf{w} \in W$ . A hyperbolic pair is a pair  $(\mathbf{e}, \mathbf{f})$  with  $q(\mathbf{e}) = q(\mathbf{f}) = 0$  and  $\beta(\mathbf{e}, \mathbf{f}) = 1$ .
- Incorrect. Non-degeneracy says  $\text{rad}(\beta) = \{\mathbf{0}\}$ , a statement about pairing with **every** vector. On  $\mathbb{R}^2$  with  $q = x_1^2 - x_2^2$  the form is non-degenerate, since its Gram matrix has determinant  $-1$ , and  $(1, 1)$  is isotropic.
- 0 for a positive definite form, since it is anisotropic, so the only totally isotropic subspace is  $\{\mathbf{0}\}$ . For signature  $(4, 4)$  the index is  $\min(4, 4) = 4$  by [Proposition 13.7.6](#) (a), the largest value [Proposition 13.7.1](#) permits in dimension 8.
- It is  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ , of determinant  $-1 \neq 0$ , and a form is non-degenerate exactly when its Gram matrix is invertible ([Proposition 13.1.2](#)).
- Non-degeneracy of  $\beta$ . Here  $\mathbf{e}_2 \in \text{rad}(\beta)$ , so there is no  $\mathbf{u}$  with  $\beta(\mathbf{e}_2, \mathbf{u}) \neq 0$  and the first line of the proof has nothing to choose.

### 13.7.7.2 B. Practice

**Exercise B1: Find the isotropic vectors.**

For each of the following real quadratic forms, find all isotropic vectors, give the signature, and state the Witt index.

- (a)  $q(x_1, x_2) = x_1^2 + 2x_1x_2$  on  $\mathbb{R}^2$ .
- (b)  $q(x_1, x_2, x_3) = x_1x_2$  on  $\mathbb{R}^3$ . (*Is this form non-degenerate?*)
- (c)  $q(x_1, x_2, x_3) = x_1^2 + x_2^2 + x_3^2$  on  $\mathbb{R}^3$ .

*Solution.*

- (a)  $q = x_1(x_1 + 2x_2)$ , so  $q = 0$  exactly on the two lines  $x_1 = 0$  and  $x_1 = -2x_2$ . The isotropic vectors are the non-zero multiples of  $(0, 1)$  and of  $(-2, 1)$ . The Gram matrix is  $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ , of determinant  $-1 < 0$ , so the eigenvalues have opposite signs and the signature is  $(1, 1)$ . The Witt index is  $\min(1, 1) = 1$ ; indeed  $(0, 1)$  is isotropic and by [Lemma 13.7.3](#) the whole space is one hyperbolic plane.
- (b) The Gram matrix is  $\begin{pmatrix} 0 & 1/2 & 0 \\ 1/2 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ , of rank 2, so the form is **degenerate** with  $\text{rad}(\beta) = \text{span}(\mathbf{e}_3)$  and the notion of Witt index does not apply as defined. The isotropic vectors are those with  $x_1x_2 = 0$ , namely the union of the two planes  $x_1 = 0$  and  $x_2 = 0$ , minus  $\mathbf{0}$ . On the non-degenerate part  $\text{span}(\mathbf{e}_1, \mathbf{e}_2)$  the signature is  $(1, 1)$  and the index is 1.
- (c) There are none:  $q$  is positive definite, so  $q(\mathbf{x}) = 0$  forces  $\mathbf{x} = \mathbf{0}$ . The signature is  $(3, 0)$  and the Witt index is 0.

**Exercise B2: Build a hyperbolic basis.**

- (a) For  $q(x_1, x_2) = x_1^2 + x_2^2$  on  $\mathbb{C}^2$ , find a hyperbolic pair.
- (b) For  $q(x_1, x_2, x_3) = x_1^2 + x_2^2 - x_3^2$  on  $\mathbb{R}^3$ , find a hyperbolic pair  $(\mathbf{e}, \mathbf{f})$  with  $\mathbf{e} = \mathbf{e}_1 + \mathbf{e}_3$ , and identify the anisotropic part  $\text{span}(\mathbf{e}, \mathbf{f})^{\perp_\beta}$ .

*Solution.*

- (a) Take  $\mathbf{e} = (1, i)$ , so  $q(\mathbf{e}) = 1 + i^2 = 0$ . The vector  $\mathbf{u} = (1, -i)$  has  $\beta(\mathbf{e}, \mathbf{u}) = 1 \cdot 1 + i \cdot (-i) = 2$ , so rescale:  $\mathbf{f} = \frac{1}{2}(1, -i)$  gives  $\beta(\mathbf{e}, \mathbf{f}) = 1$  and  $q(\mathbf{f}) = \frac{1}{4}(1 + i^2) = 0$ . So  $((1, i), \frac{1}{2}(1, -i))$  is a hyperbolic pair, and  $\mathbb{C}^2$  is a single hyperbolic plane.
- (b)  $q(\mathbf{e}) = 1 - 1 = 0$ , so  $\mathbf{e}$  is isotropic. Follow the proof of [Lemma 13.7.3](#):  $\mathbf{u}_0 = \mathbf{e}_1$  gives  $\beta(\mathbf{e}, \mathbf{e}_1) = 1$ , so take  $\mathbf{u} = \mathbf{e}_1$ , and  $q(\mathbf{u}) = 1$ , whence

$$\mathbf{f} = \mathbf{e}_1 - \frac{1}{2}(\mathbf{e}_1 + \mathbf{e}_3) = \frac{1}{2}(\mathbf{e}_1 - \mathbf{e}_3).$$

Check:  $q(\mathbf{f}) = \frac{1}{4}(1 - 1) = 0$  and  $\beta(\mathbf{e}, \mathbf{f}) = \frac{1}{2}(1 + 1) = 1$ . For the complement,  $\mathbf{v} = (v_1, v_2, v_3)$  is orthogonal to both  $\mathbf{e}$  and  $\mathbf{f}$  exactly when  $v_1 - v_3 = 0$  and  $v_1 + v_3 = 0$ , that is  $v_1 = v_3 = 0$ . So  $\text{span}(\mathbf{e}, \mathbf{f})^{\perp_\beta} = \text{span}(\mathbf{e}_2)$ , on which  $q = x_2^2$  is positive definite. This is the decomposition of [Theorem 13.7.4](#) with  $m = 1$ , matching  $\min(2, 1) = 1$ .

**Exercise B3: A totally isotropic subspace of the largest size.**

Let  $q(\mathbf{x}) = x_1x_4 + x_2x_3$  on  $\mathbb{R}^4$ .

- (a) Write down the Gram matrix and verify that  $\beta$  is non-degenerate.
- (b) Exhibit a totally isotropic subspace of dimension 2, and explain why none of dimension 3 exists.
- (c) Determine the signature of  $q$  and confirm the Witt index by [Proposition 13.7.6](#) (a).

*Solution.*

- (a) Halving the cross-term coefficients (Eq. 13.3.1),

$$\mathbf{A} = \frac{1}{2} \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix},$$

whose determinant is  $\frac{1}{16}$  det of the reversal permutation matrix, namely  $\frac{1}{16} \cdot 1 \neq 0$ . So  $\beta$  is non-degenerate by Proposition 13.1.2.

- (b) Take  $W = \text{span}(\mathbf{e}_1, \mathbf{e}_2)$ . Every  $\mathbf{w} = (a, b, 0, 0)$  has  $q(\mathbf{w}) = 0$ , and  $\beta(\mathbf{e}_1, \mathbf{e}_2) = 0$  from the matrix, so  $W$  is totally isotropic of dimension 2. No totally isotropic subspace has dimension 3, because [Proposition 13.7.1](#) caps it at  $4/2 = 2$ .
- (c) The two planes  $\text{span}(\mathbf{e}_1, \mathbf{e}_4)$  and  $\text{span}(\mathbf{e}_2, \mathbf{e}_3)$  are orthogonal, and  $(\mathbf{e}_1, 2\mathbf{e}_4)$  is a hyperbolic pair, since  $q(\mathbf{e}_1) = q(2\mathbf{e}_4) = 0$  and  $\beta(\mathbf{e}_1, 2\mathbf{e}_4) = 2 \cdot \frac{1}{2} = 1$ ; the same works for  $(\mathbf{e}_2, 2\mathbf{e}_3)$ . So each plane is hyperbolic, of signature  $(1, 1)$ . Hence  $q$  has signature  $(2, 2)$  and, by [Proposition 13.7.6](#) (a), Witt index  $\min(2, 2) = 2$ , attained by the  $W$  of part (b).

**13.7.7.3 C. Going deeper****Exercise C1: Isotropy makes the form surjective.**

Let  $\beta$  be a non-degenerate symmetric form on  $V$  over a field  $F$  with  $\text{char } F \neq 2$ .

- (a) Prove that if  $\beta$  has an isotropic vector, then  $q: V \rightarrow F$  is surjective.
- (b) Deduce that if some  $c \in F$  is not of the form  $q(\mathbf{v})$ , then  $\beta$  is anisotropic.
- (c) Use (b) to show that the form  $q(x_1, x_2) = x_1^2 + x_2^2$  on  $\mathbb{F}_3^2$  is anisotropic, and give its Witt index.

*Solution.*

- (a) Let  $\mathbf{e}$  be isotropic. By [Lemma 13.7.3](#) there is  $\mathbf{f}$  with  $(\mathbf{e}, \mathbf{f})$  a hyperbolic pair. For  $c \in F$ , put  $\mathbf{v} = \frac{c}{2}\mathbf{e} + \mathbf{f}$ ; this is legitimate since  $\text{char } F \neq 2$ . Then

$$q(\mathbf{v}) = \frac{c^2}{4}q(\mathbf{e}) + 2 \cdot \frac{c}{2}\beta(\mathbf{e}, \mathbf{f}) + q(\mathbf{f}) = 0 + c + 0 = c.$$

So every  $c \in F$  is a value of  $q$ .

- (b) Contrapositive of (a): if  $q$  is not surjective then  $\beta$  has no isotropic vector.

- (c) The squares in  $\mathbb{F}_3$  are  $0^2 = 0$ ,  $1^2 = 1$ ,  $2^2 = 1$ , so the values of  $x_1^2 + x_2^2$  are 0, 1, 2 — which is all of  $\mathbb{F}_3$ , so (b) gives nothing directly. Argue instead from the definition:  $q(\mathbf{x}) = 0$  needs  $x_1^2 = -x_2^2 = 2x_2^2$ . If  $x_2 \neq 0$  then  $x_2^2 = 1$ , so  $x_1^2 = 2$ , which is not a square in  $\mathbb{F}_3$ ; hence  $x_2 = 0$  and then  $x_1 = 0$ . The form is anisotropic, its Witt index is 0, and the Gram matrix  $\mathbf{I}_2$  shows it is non-degenerate. Over  $\mathbb{F}_5$  the same form is isotropic, since  $1^2 + 2^2 = 0$ .

**Exercise C2: A maximal totally isotropic subspace and its complement.**

Let  $\beta$  be non-degenerate symmetric on  $V$  with  $\dim V = n$ , and let  $W \subseteq V$  be totally isotropic with  $\dim W = \text{ind}(\beta) = m$ .

- (a) Prove that  $\dim W^{\perp\beta} = n - m$  and  $W \subseteq W^{\perp\beta}$ .
- (b) Prove that the form induced by  $\beta$  on the quotient  $W^{\perp\beta}/W$  is well defined and anisotropic. *Hint: a coset with an isotropic representative would enlarge  $W$ .*
- (c) Deduce that  $n - 2m = \dim(W^{\perp\beta}/W)$  agrees with the dimension of the anisotropic part in [Theorem 13.7.4](#).

*Solution.*

- (a) The dimension is Exercise (C2: Orthogonal complements for a form) (b), using non-degeneracy, and the inclusion is [Definition 13.7.2](#).
- (b) For  $\mathbf{u}, \mathbf{v} \in W^{\perp\beta}$  and  $\mathbf{w}, \mathbf{w}' \in W$ ,

$$\beta(\mathbf{u} + \mathbf{w}, \mathbf{v} + \mathbf{w}') = \beta(\mathbf{u}, \mathbf{v}),$$

because each of the three remaining terms pairs a vector of  $W$  with a vector of  $W^{\perp\beta} \supseteq W$ , hence vanishes. So  $\tilde{\beta}(\mathbf{u} + W, \mathbf{v} + W) := \beta(\mathbf{u}, \mathbf{v})$  is well defined, and it is bilinear and symmetric. Suppose  $\tilde{q}(\mathbf{u} + W) = 0$  with  $\mathbf{u} \notin W$ , that is  $q(\mathbf{u}) = 0$  with  $\mathbf{u} \in W^{\perp\beta} \setminus W$ . Then  $W + \text{span}(\mathbf{u})$  is totally isotropic: it is spanned by isotropic vectors,  $\beta(\mathbf{u}, \mathbf{w}) = 0$  for  $\mathbf{w} \in W$  because  $\mathbf{u} \in W^{\perp\beta}$ , and  $\beta$  vanishes on  $W \times W$ . Its dimension is  $m + 1 > \text{ind}(\beta)$ , a contradiction. So  $\tilde{\beta}$  is anisotropic.

- (c) By (a),  $\dim(W^{\perp\beta}/W) = (n - m) - m = n - 2m$ . By [Proposition 13.7.5](#) the anisotropic part  $V_0$  of a Witt decomposition also has dimension  $n - 2m$ , since the index is  $m$ . (That the two anisotropic forms agree, and not merely their dimensions, is proved in the next section.)

**Exercise C3: Where characteristic 2 breaks.**

Work over  $\mathbb{F}_2$  and let  $\beta$  be the dot product on  $\mathbb{F}_2^2$ , with Gram matrix  $\mathbf{I}_2$ .

- (a) Verify that  $\beta$  is symmetric and non-degenerate, and find all  $\mathbf{v}$  with  $q(\mathbf{v}) = 0$ .
- (b) Show that  $\beta$  has a vector with  $q(\mathbf{v}) = 0$  and  $\mathbf{v} \neq \mathbf{0}$  that lies in no hyperbolic plane, and identify which line of the proof of [Lemma 13.7.3](#) fails.
- (c) Explain in one sentence why the third description of “totally isotropic” given after [Definition 13.7.2](#) is unavailable here.

*Solution.*

- (a) The Gram matrix  $\mathbf{I}_2$  is symmetric and invertible, so  $\beta$  is symmetric and non-degenerate (Proposition 13.1.2). Here  $q(\mathbf{x}) = x_1^2 + x_2^2 = (x_1 + x_2)^2$  over  $\mathbb{F}_2$ , which vanishes exactly when  $x_1 = x_2$ . So  $q(\mathbf{v}) = 0$  for  $\mathbf{v} = \mathbf{0}$  and for  $\mathbf{v} = (1, 1)$ .
- (b) Take  $\mathbf{v} = (1, 1)$ . A hyperbolic plane containing it would need  $\mathbf{f}$  with  $q(\mathbf{f}) = 0$  and  $\beta(\mathbf{v}, \mathbf{f}) = 1$ . The only non-zero vector with  $q = 0$  is  $\mathbf{v}$  itself, and  $\beta(\mathbf{v}, \mathbf{v}) = q(\mathbf{v}) = 0 \neq 1$ . The step that fails is the correction  $\mathbf{f} = \mathbf{u} - \frac{1}{2}q(\mathbf{u})\mathbf{e}$ : the element  $2 = 1 + 1$  is 0 in  $\mathbb{F}_2$  and is not invertible, so  $\frac{1}{2}$  does not exist.
- (c) Because it is Theorem 13.3.5, which needs  $\text{char } F \neq 2$  to divide by 2; over  $\mathbb{F}_2$  the line  $\text{span}((1, 1))$  has  $q \equiv 0$  on it and is indeed totally isotropic here, but the implication “ $q$  vanishes on  $W$  implies  $\beta$  vanishes on  $W \times W$ ” is not available in general.